PROJECT 1 | COMPLETE POWER BI CHANGE LOG

Data Cleaning & Preparation | DecodeLabs Data Analytics Internship | 19 October 2026

1. Project purpose and scope

The goal of Project 1 is to prepare an order dataset for reliable analysis by identifying and handling missing values, checking duplicates, correcting data formats, and validating data integrity. The work was carried out using Power BI Desktop and Power Query Editor through the user interface, with one custom formula for the price-difference check.

Item	Project information
Source	Dataset for Data Analytics.xlsx, worksheet Sheet1
Initial size	1,200 data rows and 14 source columns
Working queries	Raw_Data (preserved, load disabled) and Cleaned_Data (edited and loaded)
Saved project	DecodeLabs_Project1_Data_Cleaning.pbix
Data protection	Raw source query retained; all cleaning operations applied to Cleaned_Data

2. Full sequence of work performed

Step	Power BI / Power Query action	Outcome or purpose
01	Home > Get Data > Excel workbook > Sheet1 > Transform Data	Imported the supplied workbook for cleaning rather than loading it directly.
02	Renamed Sheet1 to Raw_Data; duplicated it to Cleaned_Data; disabled Enable load on Raw_Data	Retained a query-level copy of the original and established an editable working copy.
03	View > Data view: Column quality, Column distribution, Column profile and Details pane; profiling of entire dataset	Inspected all 1,200 rows, not just the first 1,000, for empty values, errors and distinct values.
04	Selected CouponCode; inspected profile; Transform > Replace Values	Detected 309 missing codes and replaced actual nulls with No Coupon.
05	Changed Date column from numeric Excel serial format to Date data type	Made dates readable; verified 100% valid with no blanks or conversion errors.
06	Selected OrderID and inspected Column statistics	Confirmed 1,200 distinct and 1,200 unique IDs, zero nulls and zero duplicates.
07	Transform > Format > Trim on Product	Standardized outer whitespace; seven product categories remained.
08	Transform > Format > Trim on PaymentMethod, OrderStatus and ReferralSource	Standardized outer whitespace; each column retained five distinct categories.
09	Inspected Quantity, UnitPrice, ItemsInCart and TotalPrice column profiles	Verified types, ranges, missing values and errors without altering the recorded numbers.
10	Selected Quantity + UnitPrice; Add Column > Standard > Multiply; renamed Multiplication to CalculatedTotal	Created an audit calculation for each order.
11	Add Column > Custom Column: Number.Round([TotalPrice] - [CalculatedTotal], 2); named PriceDifference; set Decimal Number	Computed a two-decimal price reconciliation; min and max both 0.
12	Home > Close & Apply; File > Save As	Loaded Cleaned_Data and saved the Power BI project.

CLEANING DECISIONS & RESULTS

3. Transformation log

ID	Field / action	Recorded result and reason
CR001	CouponCode: replace null with No Coupon	309 null cells were relabeled, preserving the original 1,200 orders. This is an analytical label for missing recorded coupon codes, not proof that no coupon was used.
CR002	Date: numeric serial to Date	Converted Excel serial values such as 44930 into readable dates. Power Query reported 100% valid, 0 empty and 0 errors after conversion.
CR003	Product: Trim	Removed any leading or trailing whitespace. Seven categories remained: Chair, Desk, Laptop, Monitor, Phone, Printer and Tablet.
CR004	PaymentMethod, OrderStatus and ReferralSource: Trim	Removed outer whitespace; each had five distinct categories and 0 empty/error values after the operation.
CR005	CalculatedTotal audit column	Created using Quantity multiplied by UnitPrice. This supports reconciliation and does not overwrite TotalPrice.
CR006	PriceDifference audit column	Created using rounded difference between TotalPrice and CalculatedTotal. All values were zero to two decimal places.

4. Verification and quality-control results

Area checked	Result verified in Power Query
Dataset size	1,200 rows; 14 original source columns. Two audit columns added in Cleaned_Data.
CouponCode	309 missing values before replacement (25.75%); 0 missing after; 891 recorded codes originally.
OrderID	Count 1,200; distinct 1,200; unique 1,200; duplicate IDs 0. No rows were deleted for duplicate IDs.
Date	Date data type; 100% valid; 0 errors; 0 missing.
Quantity	Whole number; min 1; max 5; 0 missing/errors.
UnitPrice	Decimal number; min 11.39; max 699.93; mean 356.41; 0 missing/errors.
ItemsInCart	Whole number; min 1; max 10; mean 5.485; 0 missing/errors.
TotalPrice	Decimal number; min 11.39; max 3,456.40; mean 1,053.97; 0 missing/errors.
PriceDifference	Decimal number; min 0; max 0; mean 0; one distinct value (0); no nulls/errors.

5. Decisions not to modify valid data

- No duplicate removal was carried out because the OrderID profile already showed 1,200 unique IDs.
- No numeric values were changed: the checked ranges were positive, no errors or nulls were observed, and order-total reconciliation passed.
- Quantity and ItemsInCart were not forced to match, because they may represent different business concepts.

METHOD, LIMITATIONS & HANDOFF

6. Important assumptions and limits

- Coupon interpretation: No Coupon denotes an absent recorded code. Without business rules, the blank cannot be definitively interpreted as proof that a coupon was not used.
- Date interpretation: Conversion success and Power Query validity are confirmed; a separate check of whether every calendar date fits the business period was not documented.

- Price check: Equality was tested at two decimal places, consistent with recorded prices. It does not separately validate tax, shipping, or discount rules that might apply in a different dataset.
- Other fields: The inspection did not establish business-level correctness of addresses, customer identities, tracking numbers, or every categorical label.
- Evidence source: These figures are recorded from the Power Query screenshots and steps reviewed during the working session. The saved PBIX was not provided for an independent file-level audit.

7. Final query and output structure

Component	Purpose / final state
Raw_Data	Original imported sheet preserved as a reference query with Enable load turned off.
Cleaned_Data	Working dataset with null-labeling, date conversion, text trimming and calculated audit columns; loaded into Power BI.
CalculatedTotal	Quantity * UnitPrice; audit-only computed amount.
PriceDifference	Number.Round([TotalPrice] - [CalculatedTotal], 2); audit-only reconciliation.
Report canvas	No visuals required for the described data-cleaning project; empty canvas is acceptable.

8. Suggested project-folder contents

- DecodeLabs_Project1_Data_Cleaning.pbix - saved Power BI Desktop project with query transformations.
- Dataset for Data Analytics.xlsx - unmodified source data for traceability.
- DecodeLabs_Project1_Full_PowerBI_Cleaning_Log.pdf - methodology and change-log report.

9. Outcome

The recorded Power Query checks indicate that the 1,200-row order dataset was standardized for analysis without removing order records. Missing coupon codes were explicitly labeled, Excel serial dates were converted, selected categorical text columns were trimmed, duplicate OrderIDs were ruled out, numerical fields were reviewed, and order totals reconciled to two decimal places. The Power BI project was saved after Close & Apply. Submission should follow the official DecodeLabs instructions when available.

Prepared from the project instructions, source workbook and user-confirmed Power Query results.